

Regression Adjustment in Randomized Experiments: Revisiting Freedman and Lin Through Simulation

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Abstract

On regression adjustments to experimental data (Freedman, 2008), released in 2008 within *Advances in Applied Mathematics*, is a seminal paper in the field of causal inference, that challenges conventional belief that regression adjustment automatically improves casual estimation in randomized trials. Under the Neyman framework, Freedman argues that while randomization guarantees that the unadjusted difference in means is unbiased for the average treatment effect, this protection does not extend to regression models that include covariates, which can introduce bias, inflate variance. Five years later, Winston Lin addresses these concerns raised by Freedman in his paper: *Agnostic Notes on Regression Adjustments to Experimental Data* (Lin, 2013). Lin proves that adding all treatment–covariate interactions yield an estimator that is consistent, asymptotically unbiased, and never less precise than the unadjusted difference in means. This report aims to explain the key mathematical arguments of both papers, investigate the theorems with simulations, and discuss and apply the findings to broad applications.

1 Background and Theory

Assessing the effect of a treatment on a population requires comparing outcomes from this treatment against a baseline. In practice, this is typically done by dividing participants into two groups: one that receives the treatment and another that serves as a control. The gold standard for this assignment is a randomized experiment, where subjects are assigned to groups independently of their baseline characteristics. However, this begs the question: what if some of these baseline characteristics have an impact on the measured effectiveness of the treatment? How will we be able to measure the efficacy then?

This is the question that led to the concept of regression adjustment. Regression adjustment, as the name implies, allows researchers to take baseline characteristics into consideration in their regression models. The intuition is that controlling for these covariates in the model can reduce variance and improve precision. However, the statistical properties of this practice, and by extension its effectiveness, have been heavily debated in recent years. In particular, many have wondered if these adjustments have any adverse effects on model performance, such as introducing bias or increasing variance.

In 2007, Prof. David Freedman published a paper detailing certain perceived shortcomings of regression adjustment. His two main points were that multiple regression estimators, including those with covariates that were not directly affected by the treatment, were biased; however, this bias tended to zero as the sample size increased. He also showed that asymptotically, these multiple regression estimators could perform worse than a simple regression estimator, which only used the group assignment as a covariate to perform regression.

All of Freedman’s results stem from his use of the Neyman Nonparametric model. This model assumes that there are only ever 2 possible outcomes for a participant in the study:

T_i : The outcome if subject i the treatment C_i : The outcome if subject i receives the control

Only one of these is assumed to be observable in any patient, depending on which group they’re assigned to. We then measure individual treatment effect as:

$$\tau_i = T_i - C_i$$

Then, using this framework, causal inference becomes a missing data problem, as in practice T_i and C_i are never both observed at the same time. This remains true no matter which method is chosen to perform the inference, which is where the power of this model comes from. We can then define the observed response of patient i as:

$$Y_i = X_i T_i + (1 - X_i) C_i$$

Where X_i is an indicator for whether subject i was in the treatment group. When we combine this definition with the observation that in a randomized experiment, $T_i, C_i \perp\!\!\!\perp X_i$ where $\perp\!\!\!\perp$ denotes independence, we get a natural estimator for average treatment effect:

$$\tau = \mathbb{E}(T_i | X_i = 1) - \mathbb{E}(C_i | X_i = 0) = \mathbb{E}(Y_i | X_i = 1) - \mathbb{E}(Y_i | X_i = 0)$$

It's important to note in this context, T_i and C_i are not random variables but rather fixed quantities, although notably they are not necessarily the same across all subjects. That is, the Neyman model allows heterogeneity across its subjects. While this does allow the framework to be much more powerful, Freedman shows that allowing heterogeneity can lead to some problems with regression adjustment.

The goal of casual inference is to be able to estimate τ . In his paper, Freedman proposes 3 different estimators for it.

Let us assume that we have run a randomized experiment including n subjects, of which m are assigned to the treatment group and the remaining $n - m$ are assigned to the control group. Then the first estimator proposed is:

$$\hat{\tau}_{naive} = \left(\frac{1}{m} \sum_i \{Y_i : X_i = 1\} \right) - \left(\frac{1}{n - m} \sum_i \{Y_i : X_i = 0\} \right)$$

which simply corresponds to the average response in the treatment group minus the average response in the control group. He then defines: $\hat{\tau}_{SR}$, the simple regression estimator obtained by running regression of Y on X (intercept included) and $\hat{\tau}_{MR}$, the multiple regression estimator obtained by running a regression of Y on X and Z , where Z is some observable covariate not affected by treatment (intercept included).

Freedman's argument as to why regression adjustment using this framework is flawed begins by proving a few facts, namely that $\hat{\tau}_{naive} = \hat{\tau}_{SR}$ when $0 < m < n$ and that $\mathbb{E}(\hat{\tau}_{naive}) = \tau$.

To simplify upcoming proofs, we will rewrite Y_i as follows:

$$Y_i = a + \tau(X_i - p) + \delta_i$$

where:

$$p = \mathbb{E}(X_i) = \frac{m}{n} \quad \bar{T} = \frac{1}{n} \sum_{i=1}^n T_i \quad \bar{C} = \frac{1}{n} \sum_{i=1}^n C_i$$

$$a = p\bar{T} + (1 - p)\bar{C}$$

$$\alpha_i = p(T_i - \bar{T}) + (1 - p)(C_i - \bar{C}) \quad \beta_i = (T_i - \bar{T}) - (C_i - \bar{C})$$

$$\delta_i = \alpha_i + \beta_i(X_i - p)$$

The key here is to be able to expression Y_i as a regression on $X_i - p$. Centering at p does not affect estimation at all, and it will significantly simplify asymptotic analysis below. This works because we can observe $\tau = \bar{T} - \bar{C}$ exactly, which allows us to simply back to our original expression for Y_i . This also allows us to assume without loss of generality that all further covariates may be included in the model and centered accordingly.

The crux of Freedman's argument, however, is found in this theorem:

Theorem 1: Suppose:

$$\frac{1}{n} \sum_{i=1}^n \alpha_i^2 \rightarrow \overline{\alpha^2} \in \mathbb{R}, \quad \frac{1}{n} \sum_{i=1}^n \beta_i^2 \rightarrow \overline{\beta^2} \in \mathbb{R}, \quad \frac{1}{n} \sum_{i=1}^n \alpha_i \beta_i \rightarrow \overline{\alpha\beta} \in \mathbb{R} \quad (1)$$

$$\frac{1}{n} \sum_{i=1}^n \alpha_i^4 < \infty, \quad \frac{1}{n} \sum_{i=1}^n \beta_i^4 < \infty \quad (2)$$

$$0 < p < 1 \quad (3)$$

and let:

$$\gamma = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n (\alpha_i + (1-2p)\beta_i)^2 = \overline{\alpha^2} + 2(1-2p)\overline{\alpha\beta} + (1-2p)^2\overline{\beta^2} \geq 0$$

Then:

$$\sqrt{n}(\tau_{\hat{S}R} - \tau) \xrightarrow{D} N\left(0, \frac{\gamma}{p(1-p)}\right)$$

Proof: Let M denote the design matrix, i.e:

$$M = \begin{pmatrix} 1 & X_1 - p \\ 1 & X_2 - p \\ \dots & \dots \\ 1 & X_n - p \end{pmatrix}$$

Then, since $\sum_{i=1}^n X_i = np$ by construction, we get:

$$\frac{1}{n} M^T M = \begin{pmatrix} 1 & 0 \\ 0 & p(1-p) \end{pmatrix} \implies \left(\frac{1}{n} M^T M\right)^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{p(1-p)} \end{pmatrix}$$

and, if Y is the column vector of observed responses then:

$$\frac{1}{n} M^T Y = \frac{1}{n} \begin{pmatrix} \sum_{i=1}^n Y_i \\ \sum_{i=1}^n (X_i - p)Y_i \end{pmatrix}$$

$$\implies \hat{\tau}_{SR} = \frac{\sum_{i=1}^n (X_i - p)Y_i}{np(1-p)}$$

$$\implies p(1-p)\hat{\tau}_{SR} = \frac{1}{n} \sum_{i=1}^n (X_i - p)Y_i$$

$$\implies p(1-p)\hat{\tau}_{SR} = \frac{1}{n} \sum_{i=1}^n (X_i - p)(a + \tau(X_i - p) + \delta_i)$$

$$\implies p(1-p)\hat{\tau}_{SR} = \frac{a}{n} \sum_{i=1}^n (X_i - p) + \frac{\tau}{n} \sum_{i=1}^n (X_i - p)^2 + \frac{1}{n} \sum_{i=1}^n (X_i - p)\delta_i$$

We now observe:

$$i) \sum_{i=1}^n X_i = m = np \implies \sum_{i=1}^n X_i - p = np - np = 0$$

$$ii) \sum_{i=1}^n X_i^2 = np \implies \sum_{i=1}^n (X_i - p)^2 = \sum_{i=1}^n X_i^2 - 2p \sum_{i=1}^n X_i + np^2 = np - 2np^2 - np^2 = np(1-p)$$

$$\implies p(1-p)\hat{\tau}_{SR} = 0 + \frac{\tau}{n} np(1-p) + \frac{1}{n} \sum_{i=1}^n (X_i - p)\delta_i$$

$$\implies p(1-p)(\hat{\tau}_{SR} - \tau) = \frac{1}{n} \sum_{i=1}^n (X_i - p)\delta_i$$

Let $U := \frac{1}{n} \sum_{i=1}^n (X_i - p)\delta_i$. Then:

$$U = \frac{1}{n} \sum_{i=1}^n (X_i - p)\delta_i = \frac{1}{n} \sum_{i=1}^n (X_i - p)(\alpha_i + \beta_i(X_i - p)) = \frac{1}{n} \sum_{i=1}^n \alpha_i(X_i - p) + \beta_i(X_i - p)^2$$

However, since $X_i^2 = X_i$, notice:

$$\alpha_i(X_i - p) + \beta_i(X_i - p)^2 = \alpha_i X_i - \alpha_i p + \beta_i X_i^2 - \beta_i 2pX_i + \beta_i p^2 = [\alpha_i + (1 - 2p)\beta_i]X_i - \alpha_i p + \beta_i p^2$$

$$\implies U = \frac{1}{n} \sum_{i=1}^n [\alpha_i + (1 - 2p)\beta_i]X_i - \alpha_i p + \beta_i p^2$$

We now observe:

$$i) \sum_{i=1}^n \alpha_i = \sum_{i=1}^n p(T_i - \bar{T}) - (1 - p)(C_i - \bar{C}) = p(n\bar{T} - n\bar{T}) - (1 - p)(n\bar{C} - n\bar{C}) = 0$$

$$ii) \sum_{i=1}^n \beta_i = \sum_{i=1}^n (T_i - \bar{T}) - (C_i - \bar{C}) = (n\bar{T} - n\bar{T}) - (n\bar{C} - n\bar{C}) = 0$$

$$\implies U = \frac{1}{n} \sum_{i=1}^n [\alpha_i + (1 - 2p)\beta_i]X_i$$

Let $W_i = \frac{1}{\sqrt{n}}[\alpha_i + (1 - 2p)\beta_i]X_i$ for some fixed n . We notice:

$$i) \mu_i = \mathbb{E}(W_i) = \frac{1}{\sqrt{n}}[\alpha_i + (1 - 2p)\beta_i]\mathbb{E}(X_i) = \frac{1}{\sqrt{n}}p[\alpha_i + (1 - 2p)\beta_i] < \infty$$

$$ii) \sigma_i^2 = \text{Var}(W_i) = \frac{1}{n}[\alpha_i + (1 - 2p)\beta_i]^2 \text{Var}(X_i) = \frac{1}{n}p(1 - p)[\alpha_i + (1 - 2p)\beta_i]^2 < \infty$$

$$iii) W_i \perp\!\!\!\perp W_j \quad \forall i \neq j$$

Claim: Lyapunov's condition holds for W_i using $\delta = 2$, that is:

$$\lim_{n \rightarrow \infty} \frac{1}{s_n^4} \sum_{i=1}^n \mathbb{E}(|W_i - \mu_i|^4) = 0$$

where:

$$s_n^2 = \sum_{i=1}^n \sigma_i^2$$

This is because the limiting moment conditions on α_i and β_i entirely determine this sum since both μ_i and s_n are functions of alpha and beta which are limited, and X_i being Bernoulli forces its fourth moment to exist.

So then since Lyapunov's condition holds, the Lyapunov CLT will apply to the W_i 's with s_n^2 defined as above), i.e:

$$\begin{aligned} & \frac{1}{\sqrt{s_n^2}} \sum_{i=1}^n (W_i - \mu_i) \xrightarrow{D} N(0, 1) \\ \implies & \frac{1}{\sqrt{s_n^2}} \sum_{i=1}^n W_i - \frac{1}{\sqrt{s_n^2}} \sum_{i=1}^n \frac{1}{\sqrt{n}} p[\alpha_i + (1 - 2p)\beta_i] \xrightarrow{D} N(0, 1) \\ \implies & \frac{1}{\sqrt{s_n^2}} \sum_{i=1}^n W_i \xrightarrow{D} N(0, 1) \end{aligned}$$

However:

$$s_n^2 = \sum_{i=1}^n p(1 - p) \frac{1}{n} [\alpha_i + (1 - 2p)\beta_i]^2 = p(1 - p) \frac{1}{n} \sum_{i=1}^n [\alpha_i + (1 - 2p)\beta_i]^2 \rightarrow p(1 - p)\gamma$$

$$\begin{aligned}
&\implies \sqrt{n}U \xrightarrow{D} N(0, p(1-p)\gamma) \\
&\implies \sqrt{n}(p(1-p)\hat{\tau}_{SR} - \tau) \xrightarrow{D} N(0, p(1-p)\gamma) \\
&\implies \sqrt{n}(\hat{\tau}_{SR} - \tau) \xrightarrow{D} N\left(0, \frac{\gamma}{p(1-p)}\right) \quad \square
\end{aligned}$$

Through a similar logic, Freedman is able to prove that:

$$\sqrt{n}(\hat{\tau}_{MR} - \tau) \xrightarrow{D} N\left(0, \frac{\gamma'}{p(1-p)}\right)$$

where:

$$\gamma' = \gamma - \overline{\alpha Z}^2 - 2(1-2p)\overline{\alpha Z}\overline{\beta Z}$$

with $\overline{\alpha Z}$ and $\overline{\beta Z}$ defined as $\lim_{n \rightarrow \infty} \sum \alpha_i Z_i$, $\lim_{n \rightarrow \infty} \sum \beta_i Z_i$ respectively, adding conditions on finiteness as detailed below. (As a reminder, Z_i is an observable covariate not associated with treatment that is included in the regression).

We are thus able to measure the improvement in asymptotic variance as:

$$\frac{\gamma - \gamma'}{p(1-p)} = \overline{\alpha Z}(\overline{\alpha Z} - 2(1-2p)\overline{\beta Z})$$

We notice here if $p = \frac{1}{2}$, then clearly adjustment helps as in this case the improvement is $\overline{\alpha Z}^2 \geq 0$ and if there is homogeneity in treatment effects then there is no difference (as this forces $\alpha_i = \beta_i = 0 \forall i$). However, if $p \neq \frac{1}{2}$ and we have heterogeneity, then we may pick specific values for T_i and C_i so that $\overline{\alpha Z} > 0$ and $\overline{\beta Z}$ are very large. Then in this case, adjustment may hurt asymptotic variance. There are many more arguments like this to be made, essentially saying that the effect adjustment has on asymptotic precision depends on the experiment and whether or not heterogeneity is present.

However, in 2013, Prof. Winston Lin published a critique of this result and showed that if proper care is taken to set up a regression model including adjustment then the asymptotic variance of that model is not worse than the asymptotic variance of the simple model. He does this by specifying a fourth estimator for τ , $\hat{\tau}_I$, the coefficient of X_i in a regression model which contains an interaction term between X_i and Z_i .

To simplify further proofs, we once again define a few new quantities. We let:

$$\begin{aligned}
Q_T &= \left(\sum_{i=1}^n (Z_i - \overline{Z})^2 \right)^{-1} \sum_{i=1}^n (T_i - \overline{T})(Z_i - \overline{Z}) \\
Q_C &= \left(\sum_{i=1}^n (Z_i - \overline{Z})^2 \right)^{-1} \sum_{i=1}^n (C_i - \overline{C})(Z_i - \overline{Z})
\end{aligned}$$

be the asymptotic coefficients of running a regression of T_i and C_i on Z_i , respectively. We also define:

$$\begin{aligned}
Q_E &= (1-p)Q_T + pQ_C \\
\alpha_i^* &= (T_i - \overline{T}) - (Z_i - \overline{Z})Q_T \\
\beta_i^* &= (C_i - \overline{C}) - (Z_i - \overline{Z})Q_C \\
\delta_i^* &= (1-p)\alpha_i^* + p\beta_i^*
\end{aligned}$$

These will be helpful when relating Freedman's variance with Lin's. We now get to the key theorem in Lin's paper:

Theorem 2:

Suppose:

$$\frac{1}{n} \sum_{i=1}^n (\alpha_i^*)^2 \rightarrow \overline{\alpha^*}^2 \in \mathbb{R}, \quad \frac{1}{n} \sum_{i=1}^n (\beta_i^*)^2 \rightarrow \overline{\beta^*}^2 \in \mathbb{R}, \quad \frac{1}{n} \sum_{i=1}^n \alpha_i^* \beta_i^* \rightarrow \overline{\alpha^* \beta^*} \in \mathbb{R}, \quad (4)$$

$$\frac{1}{n} \sum_{i=1}^n \alpha_i^* Z_i \rightarrow \overline{\alpha^* \bar{Z}} \in \mathbb{R}, \quad \frac{1}{n} \sum_{i=1}^n \beta_i^* Z_i \rightarrow \overline{\beta^* \bar{Z}} \in \mathbb{R} \quad (5)$$

$$\frac{1}{n} \sum_{i=1}^n (\alpha_i^*)^4 < \infty, \quad \frac{1}{n} \sum_{i=1}^n (\beta_i^*)^4 < \infty, \quad \frac{1}{n} \sum_{i=1}^n Z_i^4 < \infty \quad (6)$$

$$0 < p < 1 \quad (7)$$

Then:

$$\sqrt{n}(\hat{\tau}_I - \tau) \xrightarrow{D} N\left(0, \frac{\gamma^*}{p(1-p)}\right)$$

where:

$$\gamma^* = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n (\delta_i^*)^2$$

We omit the proof of this theorem in favour of seeing how we can relate Freedman's original asymptotic variance with this new quantity, as it is similar to the proof of Theorem 1.

First, we notice:

$$\delta_i^* = (1-p)((T_i - \bar{T}) - (Z_i - \bar{Z})Q_T) + p((C_i - \bar{C}) - (Z_i - \bar{Z})Q_C)$$

We then define $E_i = (1-p)(T_i - \bar{T}) + p(C_i - \bar{C})$. Then:

$$\delta_i^* = E_i - (1-p)(Z_i - \bar{Z})Q_T - p(Z_i - \bar{Z})Q_C = E_i - (Z_i - \bar{Z})((1-p)Q_T + pQ_C) = E_i - (Z_i - \bar{Z})Q_E$$

$$\implies \gamma^* = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n (E_i - (Z_i - \bar{Z})Q_E)^2$$

However, we must also notice:

$$\alpha_i + (1-2p)\beta_i = p(T_i - \bar{T}) - (1-p)(C_i - \bar{C}) + (1-2p)((T_i - \bar{T}) - (C_i - \bar{C}))$$

$$\implies \alpha_i + (1-2p)\beta_i = (1-p)(T_i - \bar{T}) + p(C_i - \bar{C}) = E_i$$

$$\implies \gamma = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n E_i^2$$

So then:

$$\implies \gamma - \gamma^* = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n E_i^2 - (E_i - (Z_i - \bar{Z})Q_E)^2 \geq 0$$

This forces the asymptotic variance of the interaction model to be less than that of the unadjusted model, with equality maintained if and only if Q_E is zero, which coincides with the covariates being uncorrelated and having a weighted average:

$$\frac{n-m}{n}T_i + \frac{m}{n}C_i$$

2 Experiments

To complement the theoretical results, we carry out a simulation study that compares the finite sample behavior of three estimators of the average treatment effect: the unadjusted difference in means, the pooled OLS estimator without interactions, and the fully interacted OLS estimator. The simulations are designed to mirror the situations considered by Freedman and Lin. By varying only two features, whether treatment effects are constant or heterogeneous and whether the regression model includes an interaction between treatment and the covariate, the design isolates when regression adjustment helps, when it can hurt, and when the fully interacted specification restores good performance.

In Cases 1 and 2 treatment effects are constant. The individual treatment effect is the same for every unit,

$$\tau_i = 1 \quad \text{for all } i.$$

In Cases 3 and 4 treatment effects are heterogeneous and depend on the sign of Z_i ,

$$\tau_i = \begin{cases} 1, & Z_i \leq 0, \\ 2, & Z_i > 0. \end{cases}$$

We compare two regression specifications. The pooled model without interactions adjusts only linearly for the covariate,

$$Y_i = \alpha + \tau X_i + \beta Z_i,$$

and the interacted model allows the slope on Z_i to differ by treatment status,

$$Y_i = \alpha + \tau X_i + \beta Z_i + \gamma(X_i Z_i).$$

In Case 1 treatment effects are constant and the pooled model without interactions is used. There is no heterogeneity and no interaction term in the regression, so ordinary regression adjustment is correctly specified.

In Case 2 treatment effects are still constant, with $\tau_i = 1$ for all i , but the interacted model is used in estimation. There is no heterogeneity but the regression includes the extra $X_i Z_i$ term. This case examines whether adding an unnecessary interaction term has a meaningful cost.

In Case 3 treatment effects are heterogeneous, with τ_i equal to 1 for units with $Z_i \leq 0$ and 2 for units with $Z_i > 0$, but the pooled model without interactions is used. Here there is heterogeneity and no interaction term in the regression, so the model is misspecified. This recreates the situation in which regression adjustment can perform worse than the unadjusted difference in means.

In Case 4 treatment effects are heterogeneous in exactly the same way as in Case 3, with $\tau_i = 1$ if $Z_i \leq 0$ and 2 if $Z_i > 0$, but the interacted model is used. There is heterogeneity and the regression includes the interaction term $X_i Z_i$. This is the correctly specified interacted specification that Lin recommends and it is expected to perform well even when treatment effects vary with Z_i .

3 Results

case_id	n	mean_true	mean_unadj	bias_unadj	sd_unadj	mse_unadj	mean_pooled	bias_pooled	sd_pooled	mse_pooled	mean_inter	bias_inter	sd_inter	mse_inter
1.0000	30.0000	1.0000	1.0058	0.0058	0.5247	0.2752	1.0026	0.0026	0.3862	0.1491	1.0043	0.0043	0.3930	0.1544
1.0000	100.0000	1.0000	1.0012	0.0012	0.2849	0.0811	0.9991	-0.0009	0.2066	0.0427	0.9984	-0.0016	0.2081	0.0433
1.0000	500.0000	1.0000	0.9982	-0.0018	0.1270	0.0161	0.9977	-0.0023	0.0885	0.0078	0.9977	-0.0023	0.0885	0.0078
1.0000	1,000.0000	1.0000	0.9979	-0.0021	0.0881	0.0078	0.9983	-0.0017	0.0634	0.0040	0.9984	-0.0016	0.0635	0.0040
1.0000	5,000.0000	1.0000	1.0003	0.0003	0.0395	0.0016	1.0000	0.0000	0.0275	0.0008	1.0000	0.0000	0.0275	0.0008
2.0000	30.0000	1.0000	0.9947	-0.0053	0.5241	0.2746	0.9952	-0.0048	0.3840	0.1474	0.9940	-0.0060	0.3960	0.1568
2.0000	100.0000	1.0000	1.0042	0.0042	0.2886	0.0833	0.9981	-0.0019	0.2035	0.0414	0.9980	-0.0020	0.2053	0.0421
2.0000	500.0000	1.0000	0.9991	-0.0009	0.1268	0.0161	0.9992	-0.0008	0.0891	0.0079	0.9990	-0.0010	0.0893	0.0080
2.0000	1,000.0000	1.0000	0.9984	-0.0016	0.0876	0.0077	0.9997	-0.0003	0.0615	0.0038	0.9997	-0.0003	0.0615	0.0038
2.0000	5,000.0000	1.0000	0.9995	-0.0005	0.0396	0.0016	0.9998	-0.0002	0.0280	0.0008	0.9998	-0.0002	0.0280	0.0008
3.0000	30.0000	1.4983	1.5075	0.0092	0.4400	0.1834	1.5088	0.0105	0.4433	0.1852	1.5008	0.0024	0.5502	0.3046
3.0000	100.0000	1.4999	1.5001	0.0002	0.2423	0.0560	1.5004	0.0004	0.2377	0.0538	1.5001	0.0002	0.2332	0.0550
3.0000	500.0000	1.5005	1.5003	-0.0002	0.1060	0.0109	1.5010	0.0004	0.1034	0.0103	1.5015	0.0009	0.1011	0.0104
3.0000	1,000.0000	1.5000	1.4989	-0.0011	0.0740	0.0052	1.4989	-0.0012	0.0723	0.0050	1.4986	-0.0014	0.0703	0.0050
3.0000	5,000.0000	1.5001	1.5004	0.0003	0.0338	0.0011	1.5006	0.0005	0.0330	0.0010	1.5006	0.0005	0.0322	0.0010
4.0000	30.0000	1.5014	1.5272	0.0257	0.6797	0.4501	1.5155	0.0140	0.4508	0.1917	1.5220	0.0205	0.4757	0.2265
4.0000	100.0000	1.4988	1.5081	0.0093	0.3631	0.1286	1.5063	0.0075	0.2377	0.0537	1.5058	0.0071	0.2342	0.0557
4.0000	500.0000	1.5005	1.5011	0.0006	0.1660	0.0270	1.5013	0.0008	0.1058	0.0106	1.5010	0.0005	0.1020	0.0104
4.0000	1,000.0000	1.4991	1.4998	0.0007	0.1162	0.0133	1.4974	-0.0017	0.0729	0.0051	1.4976	-0.0015	0.0709	0.0052
4.0000	5,000.0000	1.4998	1.5003	0.0004	0.0496	0.0024	1.5006	0.0008	0.0324	0.0010	1.5008	0.0010	0.0317	0.0010

Figure 1: True average treatment effect (ATE), bias, standard deviation, and mean squared error (MSE) for each estimator, by case and sample size.

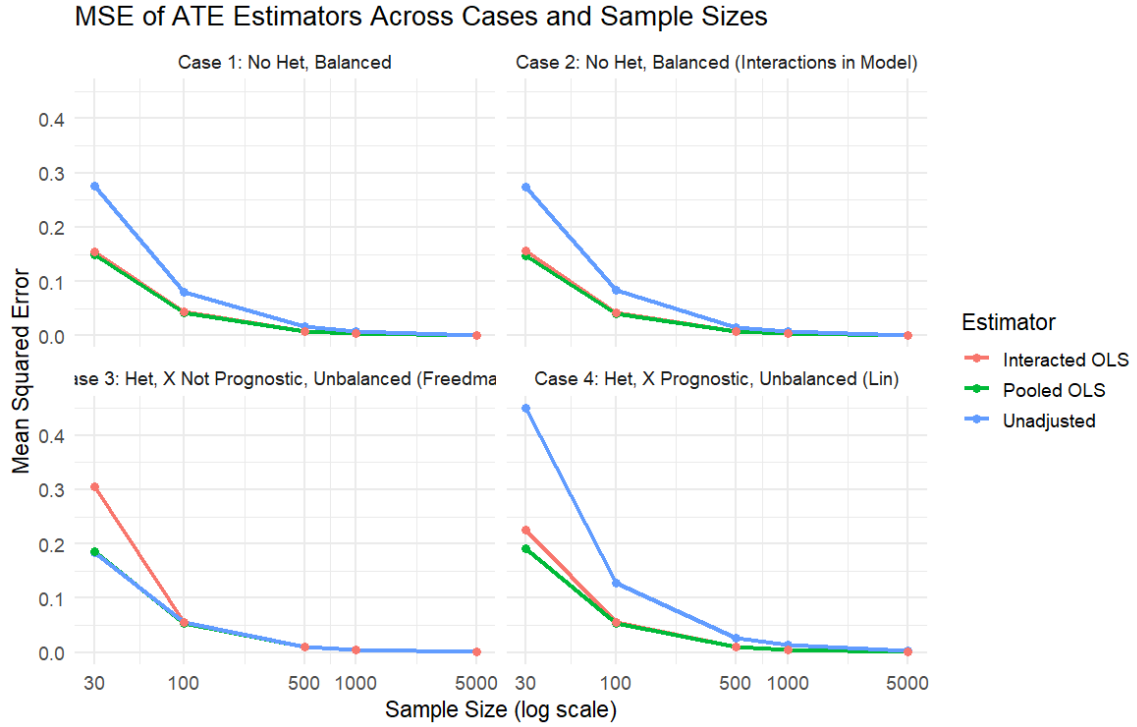


Figure 2: Mean squared error of the unadjusted, pooled OLS, and interacted OLS estimators across the four simulation cases. Each panel corresponds to one case; the x -axis shows sample size ($n \in \{30, 100, 500, 1000, 5000\}$) on a log scale.



Figure 3: Individual treatment effects τ_i as a function of the covariate Z_i in Cases 3 and 4. Units with $Z_i \leq 0$ have $\tau_i = 1$, and units with $Z_i > 0$ have $\tau_i = 2$.

Figure 1 reports summary statistics (bias, standard deviation, and mean squared error) for each estimator across all simulation scenarios. Figure 2 plots the mean squared error (MSE) of the three ATE estimators across sample sizes for each of the four cases, and Figure 3 displays the individual-level treatment effect heterogeneity imposed in Cases 3 and 4.

Across all four cases, MSE decreases substantially as the sample size increases from $n = 30$ to $n = 500$, and differences between estimators become small by $n = 1,000$ and $n = 5,000$. The clearest contrasts emerge at smaller sample sizes.

Case 1: No Heterogeneity, Balanced, Pooled Model

In Case 1, treatment effects are constant and the pooled OLS model is correctly specified. Figure 2 shows that at $n = 30$, the pooled estimator has the lowest MSE (approximately 0.15), compared with 0.27 for the unadjusted estimator. The interacted estimator is only slightly worse than pooled (MSE ≈ 0.16).

Table values in Figure 1 indicate that all estimators have negligible bias even at small n . By $n = 500$ and above, all three estimators exhibit MSE below 0.02 and are nearly indistinguishable.

Case 2: No Heterogeneity, Balanced, Interactions Included

Case 2 also has constant treatment effects, but estimation uses the interacted specification even though the interaction is unnecessary. Figure 2 shows nearly the same pattern as Case 1.

At $n = 30$, pooled OLS again achieves the lowest MSE (about 0.147), with the interacted estimator around 0.157 and the unadjusted estimator around 0.275. As shown in Figure 1, the biases remain extremely small across all sample sizes.

For $n \geq 100$, the MSE curves for pooled and interacted OLS coincide, and differences from the unadjusted estimator become minor.

Case 3: Heterogeneous Effects, Unbalanced, Z Not Prognostic (Freedman Setting)

In Case 3, treatment effects vary with the sign of Z_i , but Z_i has no prognostic value for outcomes. The pooled regression omits the interaction and is thus misspecified.

Figure 2 (bottom-left panel) and Figure 1 together show:

- At $n = 30$, unadjusted and pooled OLS have very similar MSE (approximately 0.18–0.19).
- The interacted estimator has substantially larger MSE (approximately 0.30), driven by higher variance.
- At $n = 100$, pooled OLS improves slightly over unadjusted, and the interacted estimator remains somewhat noisier.
- For $n \geq 500$, MSEs for all estimators converge to about 0.01.

Thus, in this Freedman-style setting, differences between unadjusted and pooled OLS are small and predominantly visible at the smallest sample size, while including an unnecessary interaction can introduce extra variability at small n .

Case 4: Heterogeneous Effects, Unbalanced, Z Prognostic (Lin Setting)

Case 4 features the same heterogeneous treatment effects as Case 3 (shown in Figure 3) but now Z_i is prognostic. Estimators that adjust for Z_i therefore have a chance to reduce variance.

Figure 2 (bottom-right panel) and Figure 1 show a clear advantage for regression adjustment:

- At $n = 30$, the unadjusted estimator has an MSE around 0.45.
- The pooled and interacted estimators have much smaller MSEs of about 0.19 and 0.23.
- At $n = 100$, regression adjustment continues to reduce MSE (approximately 0.054–0.056 vs. 0.13 unadjusted).
- For $n \geq 500$, pooled and interacted OLS achieve nearly identical MSEs (< 0.01), consistently outperforming the unadjusted estimator.

Thus, when Z is prognostic and treatment effects vary, regression adjustment provides a substantial finite-sample improvement.

4 Discussion & Conclusion

The results of our simulation across all four cases confirm some of the concerns of Freedman, but ultimately validate the positive clarifications suggested by Lin.

In case 1, where the τ_i were fully balanced and each set to 1, and the regression model did not include an interaction term, both the pooled and interacted OLS estimators for τ outperformed the naive unadjusted difference in means. In particular, the interacted estimator for τ did have a slightly worse MSE than the non-interacted, but the difference was small enough to be *almost* negligible. For case 2 where the regression model incorporated an interaction between the X_i and Z_i , MSE was about the same for each estimator as it had been with the non-interacted regression model. The latter case here serves as our first major point of evidence that additional interaction terms asymptotically cannot harm a model’s performance.

Case 3 is where we began to simulate a dependence of τ_i on Z_i . When using regression without interaction, at low sample sizes the interacted OLS for τ performed *significantly* worse than the pooled OLS or unadjusted estimators, the two of which performed about equally. Although the pooled OLS estimator did not underperform the unadjusted, using an adjusted estimator not appropriate to the model (namely the interacted) caused a noticeable loss in performance in the form of extra variability. This recreates a scenario in which Lin suggests that Freedman’s findings hold true, and we observe an instance of this in the form of the interacted estimator. Even so, this effect diminished asymptotically, yielding the additional insight that even in the situation which Lin explained could still cause a drop in model performance, this behaviour can be effectively erased with increasing sample size.

Case 4 turns this into a scenario modelling with interacted regression, the same type which was suggested by Lin to remove the possibility of underperformance in regression adjustment. Here, the unadjusted estimator underperformed drastically compared to the adjusted estimators, and this difference remained true as sample size increased, even beyond $n \geq 500$. When compared with the results of case 3, this confirms Lin’s suggestion that adding interaction terms to a model can increase

performance when using regression adjustments.

In particular, these observations support Lin’s ideas: many of the pitfalls suggested by Freedman can be avoided by estimators appropriate to the model at hand, and furthermore by picking an appropriate model for the scenario.

The design of our experiment allowed insight into the strength in choosing an appropriate model and subsequent estimators, but there are additional scenarios described by Freedman as capable of producing an impact on model performance.

In particular, Freedman expresses that the value of p can have an impact on performance when adjusting for covariates: if $p = \frac{1}{2}$ then adjustment cannot hurt performance, but if $p \neq \frac{1}{2}$ then adjustment opens the possibility of increasing variability. Of note, this is because the gain in adjustment is $\frac{(\bar{\alpha Z})[(\bar{\alpha Z}) + 2(1-2p)(\beta Z)]}{np(1-p)}$, and $p = \frac{1}{2} \implies (\bar{\alpha Z})^2 \geq 0$ ends up being the only term in the numerator, ensuring that the improvement is non-negative. To provide further commentary on his findings, specific values of p could be controlled for in sampling, allowing for observation of the potential decrease in performance when p is allowed to vary as opposed to when it is fixed at $\frac{1}{2}$.

Additional work may also be undertaken to observe the impact of regression adjustment outside of the Neyman model adhered to by Freedman and Lin. This would involve the use of an entirely different model and construction of the response, and so would be far more involved than the previous suggestion, but warrants investigation considering the utility in the results for this case.

To conclude, Freedman’s rigorous proof warrants consideration of misuse of regression adjustment, but Lin’s argument provides additional clauses for situations not originally considered. Specifically, adjusting the regression model for interactions in covariates and not just the covariates themselves can ensure no loss in performance when adjusting. Our simulated experiments seemed to confirm this, as estimating regressors using methods inappropriate to the model at hand resulted in reduced performance at low sample sizes compared to both an unadjusted estimator and a more appropriate adjusted estimator. Conversely, ensuring an appropriate adjusted estimator was used seemed to ensure that no significant increase in variability occurred when compared to an unadjusted estimator. But, importantly, this discrepancy disappeared asymptotically with increasing sample size. This yields a significant takeaway for researchers making use of regression methods in a randomized experiment setting: using a fully interacted regression specification protects against the pitfalls Freedman identifies, especially in large samples.

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References

- [1] Freedman, D. A. (2008). On regression adjustments to experimental data. *Advances in Applied Mathematics*, 40(2), 180–193.
- [2] Lin, W. (2013). Agnostic notes on regression adjustments to experimental data: Reexamining Freedman’s critique. *The Annals of Applied Statistics*, 7(1), 295–318.

Appendix

R code for the simulation study

```

1 <library(kableExtra)
2 library(gt)
3 library(dplyr)
4 library(tidyverse)
5
6 set.seed(123)

```

```

7
8 generate_data_case <- function(n, case_id) {
9   X <- rnorm(n, 0, 1)
10  eps <- rnorm(n, 0, 1)
11  alpha <- 0
12
13  if (case_id %in% c(1, 2)) {
14    ## cases 1 and 2 balanced no heterogeneity
15    theta <- 1
16    p_t <- 0.5
17    tau_i <- rep(1, n)
18
19    Y0 <- alpha + theta * X + eps
20    Y1 <- Y0 + tau_i
21
22  } else if (case_id == 3) {
23    ## case 3 yes heterogeneity and unbalanced
24    theta <- 0
25    p_t <- 0.3
26    tau_i <- ifelse(X > 0, 2, 1)
27
28    Y0 <- alpha + theta * X + eps
29    Y1 <- Y0 + tau_i
30
31  } else if (case_id == 4) {
32    ## case 4 yes heterogeneity and unbalanced
33    theta <- 1
34    p_t <- 0.3
35    tau_i <- ifelse(X > 0, 2, 1)
36
37    Y0 <- alpha + theta * X + eps
38    Y1 <- Y0 + tau_i
39
40  } else {
41    stop("case_id must be 1, 2, 3, or 4.")
42  }
43
44  # treatment assignment
45  T_i <- rbinom(n, size = 1, prob = p_t)
46
47  # observed outcome
48  Y <- ifelse(T_i == 1, Y1, Y0)
49
50  data.frame(
51    Y = Y,
52    T = T_i,
53    X = X,
54    tau_i = tau_i,
55    true_ATE = mean(tau_i)
56  )
57 }
58
59 ## estimator function
60
61 estimate_all <- function(dat) {
62   # Unadjusted difference in means
63   ate_unadj <- with(dat, mean(Y[T == 1]) - mean(Y[T == 0]))
64
65   # Pooled OLS: Y ~ T + X
66   fit_pooled <- lm(Y ~ T + X, data = dat)
67   ate_pooled <- coef(fit_pooled)["T"]
68
69   # Interacted OLS: Y ~ T * X
70   fit_inter <- lm(Y ~ T * X, data = dat)
71   ate_inter <- coef(fit_inter)["T"]

```

```

72
73   c(
74     true_ATE = unique(dat$true_ATE),
75     unadj     = ate_unadj,
76     pooled    = ate_pooled,
77     inter     = ate_inter
78   )
79 }
80
81 ## simulation for one (case, n) pair
82
83 run_sim_case_n <- function(case_id, n, nsim = 2000) {
84   res <- matrix(NA_real_, nrow = nsim, ncol = 4)
85   colnames(res) <- c("true_ATE", "unadj", "pooled", "inter")
86
87   for (s in 1:nsim) {
88     dat <- generate_data_case(n = n, case_id = case_id)
89     est <- estimate_all(dat)
90     res[s, ] <- est
91   }
92
93   res <- as.data.frame(res)
94
95   ## summary statistics: means, bias, sd, mse
96   mean_true <- mean(res$true_ATE)
97
98   mean_unadj <- mean(res$unadj)
99   bias_unadj <- mean(res$unadj - res$true_ATE)
100  sd_unadj <- sd(res$unadj)
101  mse_unadj <- mean((res$unadj - res$true_ATE)^2)
102
103  mean_pooled <- mean(res$pooled)
104  bias_pooled <- mean(res$pooled - res$true_ATE)
105  sd_pooled <- sd(res$pooled)
106  mse_pooled <- mean((res$pooled - res$true_ATE)^2)
107
108  mean_inter <- mean(res$inter)
109  bias_inter <- mean(res$inter - res$true_ATE)
110  sd_inter <- sd(res$inter)
111  mse_inter <- mean((res$inter - res$true_ATE)^2)
112
113  data.frame(
114    case_id      = case_id,
115    n            = n,
116    mean_true    = mean_true,
117
118    mean_unadj   = mean_unadj,
119    bias_unadj   = bias_unadj,
120    sd_unadj     = sd_unadj,
121    mse_unadj    = mse_unadj,
122
123    mean_pooled  = mean_pooled,
124    bias_pooled  = bias_pooled,
125    sd_pooled    = sd_pooled,
126    mse_pooled   = mse_pooled,
127
128    mean_inter   = mean_inter,
129    bias_inter   = bias_inter,
130    sd_inter     = sd_inter,
131    mse_inter    = mse_inter
132  )
133 }
134
135 ## run all cases and sample sizes
136

```

```

137 sample_sizes <- c(30, 100, 500, 1000, 5000)
138 nsim <- 2000
139
140 all_results_list <- list()
141 idx <- 1
142
143 for (case_id in 1:4) {
144   for (n in sample_sizes) {
145     cat("Running case", case_id, "n =", n, "\n")
146     all_results_list[[idx]] <- run_sim_case_n(case_id, n, nsim = nsim)
147     idx <- idx + 1
148   }
149 }
150
151 results_summary <- do.call(rbind, all_results_list)
152 row.names(results_summary) <- NULL
153
154 results_summary %>%
155   gt() %>%
156   tab_header(
157     title = "Simulation Results Across Cases and Sample Sizes"
158   ) %>%
159   fmt_number(
160     columns = where(is.numeric),
161     decimals = 4
162   )
163
164 mse_long <- results_summary %>%
165   select(case_id, n, mse_unadj, mse_pooled, mse_inter) %>%
166   pivot_longer(cols = starts_with("mse"),
167     names_to = "estimator",
168     values_to = "mse") %>%
169   mutate(estimator = recode(estimator,
170     "mse_unadj" = "Unadjusted",
171     "mse_pooled" = "Pooled OLS",
172     "mse_inter" = "Interacted OLS"))
173
174 ggplot(mse_long, aes(x = n, y = mse, color = estimator)) +
175   geom_line(size = 1.1) +
176   geom_point(size = 2) +
177   scale_x_log10(breaks = c(30, 100, 500, 1000, 5000)) +
178   facet_wrap(~ case_id, nrow = 2,
179     labeller = labeller(
180       case_id = c(
181         '1' = "Case 1: No Het, Balanced",
182         '2' = "Case 2: No Het, Balanced (Interactions in Model)",
183         '3' = "Case 3: Het, X Not Prognostic, Unbalanced (Freedman)",
184         '4' = "Case 4: Het, X Prognostic, Unbalanced (Lin)"
185       )
186     )) +
187   labs(
188     title = "MSE of ATE Estimators Across Cases and Sample Sizes",
189     x = "Sample Size (log scale)",
190     y = "Mean Squared Error",
191     color = "Estimator"
192   ) +
193   theme_minimal(base_size = 13)
194
195 dat_ex <- generate_data_case(n = 500, case_id = 3) # or case 4
196
197 ggplot(dat_ex, aes(x = X, y = tau_i)) +
198   geom_point(alpha = 0.4, size = 2) +
199   geom_vline(xintercept = 0, linetype = "dashed", color = "red") +
200   labs(

```

```
201     title = "True Treatment Effect Heterogeneity in Cases 3 and 4",
202     x = "Covariate X",
203     y = "Individual Treatment Effect"
204 ) +
205 theme_minimal(base_size = 14)
206 >
```